

4. Limits

4.4. L'Hospital's Rule

4.4a Learning Objectives

- Identify the different types of indeterminate forms a limit can take
- Use L'Hospital's Rule for indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$
- Use techniques to rearrange limits of other indeterminate forms in order to use L'Hospital's Rule

Note

We are going on a tiny detour here back to chapter 4 and limit evaluation techniques so that we can build up some tools we will need in future sections. Specifically, we want to be able to evaluate integrals with infinity in the bounds or a vertical asymptote in the integrand (see section 7.8). Both of these techniques require us to evaluate limits. SO, we will spend 2 days going over L'Hospital's rule to help us with these limits.

4.4.1. Motivation

Find the following limit: $\lim_{x \rightarrow \infty} \frac{\ln|x|}{x^2}$

Thinking back to limit rules, we can consider a limit of a ration as the ratio of the limits so long as they exist (and therefore are finite). Checking if this is possible, we see

$$\lim_{x \rightarrow \infty} \ln(|x|) = \infty$$

and

$$\lim_{x \rightarrow \infty} x^2 = \infty$$

Neither of these limits exist, so we cannot use this good old limit rule, but if we check the graphs of these functions, it looks like the denominator grows much faster than the numerator. It sort of feels like the limit of the entire fraction should exist!

Thinking a bit more about how the function "looks like the denominator grows much faster than the numerator" we may start to think about derivatives. More on this in a moment.

Lastly, if we have a limit that looks like $\frac{\infty}{\infty}$ it could go to a few different values. In math, there are different sizes of infinity, but we don't really have a different way to write them (we kind of do but unless you're a math major we don't need to get into this). If the infinity on top of the fraction is larger, then the overall limit could be infinite; on bottom, then the overall limit could be 0 (and we also have to think about if they are approximately the same). There are other operations we can apply to ∞ that are similarly **indeterminate** depending on the relative sizes of the infinities.

Warning

Recall that when we take limits that go to ∞ we get a bit lazy with notation, ∞ is not a number so we cannot say it is equal to anything. However, when we say $\lim_{x \rightarrow \infty} x^2 = \infty$ what we mean is "as $x \rightarrow \infty$, $x^2 \rightarrow \infty$ ". When we discuss indeterminate forms be very careful to never say "something = $\frac{\infty}{\infty}$ " (or other indeterminate forms).

Important

4.4.2. Types of Indeterminate Forms of Limits

The following are indeterminate forms for limits. If you get any of these when trying to evaluate a limit, you cannot conclude anything about the limit you want to take. Further action must be tried.

$$\begin{array}{cccc}\frac{\pm\infty}{\pm\infty} & \frac{0}{0} & \infty - \infty & \\ \pm\infty \cdot 0 & 1^\infty & 0^0 & \infty^0\end{array}$$

The following are NOT indeterminate forms.

- $-\infty - \infty$: this goes to $-\infty$
- $\infty \cdot \infty$: this goes to ∞
- $0 \cdot 0$: this is still equal to 0, just like when you learned multiplication the first time

Theorem 4.4.1 L'Hospital's Rule

Suppose that f and g are differentiable on an open interval I containing a , and that $g'(x) \neq 0$ on I if $x \neq a$. If either

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$$

or

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \pm\infty$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

assuming that the limit on the right side of this equation exists.

Remark

- While l'Hospital first published the calculus textbook with this result, it was his tutor, Johann Bernoulli, who first discovered it. L'Hospital was paying him, and so we still refer to it by his name, however. Also note, if you get far enough in any math or science class you may hear a lot about the Bernoulli Family!
- The spelling of l'Hospital is more along the lines of how he would have spelt his name back in the 1600s, however you may see this name spelt in a more modern way as l'Hôpital in other sources.

Warning

After learning about l'Hospital's rule, some students forget the quotient rule for derivatives exists, and/or forget regular limit evaluation techniques. Take a moment to review those again and write out the differences between these and l'Hospital's rule. When do you use each, and how are they

used? See sections 2.3 (Limit Laws) and 3.2 (Quotient Rule) if you feel you may need to review these distinctions.

4.4.3. Limit Examples

For each of the following examples, evaluate the limit if it exists. If it does not exist, be as specific as possible. If using l'Hospital's rule, make sure to show which indeterminate form the limit is in.

Example 4.4.1

$$\lim_{x \rightarrow \infty} \frac{\ln|x|}{x^2}$$

Solution

We have already started to discuss this limit, and so we know it has indeterminate form $\frac{\infty}{\infty}$. Another way to write this sentence is

$$\lim_{x \rightarrow \infty} \frac{\ln|x|}{x^2} \text{ has indeterminate form } \frac{\infty}{\infty}.$$

Then,

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\ln|x|}{x^2} &\stackrel{\text{LH}}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{2x} \\ &= \lim_{x \rightarrow \infty} \frac{1}{2x^2} \\ &= 0 \end{aligned}$$

Warning

$\frac{\infty}{\infty}$ is not a number. If you write $\lim_{x \rightarrow \infty} \frac{\ln|x|}{x^2} = \frac{\infty}{\infty}$, you are writing nonsense, and you will lose points over it.

Example 4.4.2

$$\lim_{x \rightarrow \pi^-} \frac{\sin(x)}{1 - \cos(x)}$$

Solution

Our first limit evaluation technique is always just to check direct substitution...

$$\lim_{x \rightarrow \pi^-} \frac{\sin(x)}{1 - \cos(x)} = \frac{\sin(\pi)}{1 - \cos(\pi)} = \frac{0}{2} = 0$$

L'Hospital's rule was not needed.

Remark

What if we did do l'Hospital's rule in the above problem anyway? Would it still work?

$$\begin{aligned}\lim_{x \rightarrow \pi^-} \frac{\sin(x)}{1 - \cos(x)} &\stackrel{\text{LH}}{=} \lim_{x \rightarrow \pi^-} \frac{\cos(x)}{\sin(x)} \\ &= \frac{-1}{0^+} \\ &= -\infty \neq 0\end{aligned}$$

If we used l'Hospital's rule when not in indeterminate form, it gives us the wrong answer! This is because this function was not in indeterminate form.

Example 4.4.3

$$\lim_{x \rightarrow 0^+} \frac{x + \sin(x)}{x}$$

Solution

$\lim_{x \rightarrow 0^+} \frac{x + \sin(x)}{x}$ has indeterminate form of $\frac{0}{0}$, so we can use l'Hospital's rule. Then,

$$\begin{aligned}\lim_{x \rightarrow 0^+} \frac{x + \sin(x)}{x} &\stackrel{\text{LH}}{=} \lim_{x \rightarrow 0^+} \frac{1 + \cos(x)}{1} \\ &= \frac{1 + 1}{1} = 2\end{aligned}$$

Example 4.4.4

$$\lim_{x \rightarrow 0^+} \sin(x) \ln(x)$$

Solution

$\lim_{x \rightarrow 0^+} \sin(x) \ln(x)$ has indeterminate form of $0 \cdot \infty$, so it is unclear what this limit is, however l'Hospital's rule only applies to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ cases. So, we must first rewrite this one!

$$\begin{aligned}\lim_{x \rightarrow 0^+} \sin(x) \ln(x) &= \lim_{x \rightarrow 0^+} \frac{\ln(x)}{\csc(x)} \text{ which has indeterminate form } \frac{\infty}{\infty} \\ &\stackrel{\text{LH}}{=} \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\csc(x) \cot(x)} \\ &= \lim_{x \rightarrow 0^+} \frac{-\sin(x) \tan(x)}{x} \text{ which has indeterminate form } \frac{0}{0} \\ &\stackrel{\text{LH}}{=} \lim_{x \rightarrow 0^+} -\sin(x) \sec^2(x) - \cos(x) \tan(x) \\ &= -\sin(0) \sec^2(0) - \cos(0) \tan(0) \\ &= (0)(1) - (1)(0) = 0\end{aligned}$$

Notice that we used l'Hospital's rule twice, and we labeled when we did so each step, and which indeterminate form we had too. It is really hard to follow without these notes, so they are essentially required.

Tip

Sometimes we need to rewrite a function using

$$f(x) = \frac{1}{\frac{1}{f(x)}}$$

to help us get from an indeterminate form we can't use l'Hospital's rule to one we can ($\frac{0}{0}$ or $\frac{\infty}{\infty}$)

4.4a Section Summary:

- We listed some indeterminate forms for limits (IFL).
- We discussed when we can use l'Hospital's rule to evaluate limits.
- We evaluated limits, some with l'Hospital's rule but not all.
- We rewrote limits using algebra to get to an indeterminate form that we can use l'Hospital's rule on.